

Final Exam R1

1. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $v \in \mathbb{R}^n$. If f is differentiable at $x \in \mathbb{R}^n$, prove that the directional derivative is $Df_v(x) = \nabla f(x) \cdot v$. **1p**
2. Consider the curve defined by the equation $f(x, y) = 20$, where $f(x, y) = x^2 - 2x + y^2 - 4y$.
 - (a) Find the direction in which the function f increases the most at the point $(4, 6)$. **0.5p**
 - (b) Write a gradient descent method to find the minimum of f , starting from $(4, 6)$. **0.5p**
 - (c) Find the equation of the tangent line to the curve at the point $(4, 6)$. **1p**
3. Consider the function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$, $f(x, y, z) = x^2y + yz^2 - xz + 2y^2$.
 - (a) How many critical points does the function f have? **1p**
 - (b) Classify the critical point $(0, 0, 0)$. **1p**
 - (c) Write the second order Taylor polynomial for f centered around $(0, 0, 0)$. **1p**
4. Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & 3 \end{bmatrix}$, $b = \begin{bmatrix} 8 \\ 28 \end{bmatrix}$, $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, $\alpha > 0$. Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$, $f(x) = \|x\|^2 + \alpha \|Ax - b\|^2$.
 - (a) Find the unique $x^* \in \mathbb{R}^3$ that solves the problem: $\min_{x \in \mathbb{R}^3} \|x\|^2$ subject to $Ax = b$. **1p**
 - (b) Find the unique $x_\alpha^* \in \mathbb{R}^3$ that minimizes f over $\mathbb{R}^3$. Show that $\lim_{\alpha \rightarrow \infty} x_\alpha^* = x^*$. **1p**
5. Find $\iint_D (1 + x^2 + y^2)^{3/2} dx dy$, where $D = \{(x, y) \in \mathbb{R}^2 \mid 1 \leq x^2 + y^2 \leq 4\}$. **2p**

Time: 2h. Total number of points: 10p. Please write your full name, your group and your row number R1.